The third boundary coefficient of even off-diagonally symmetric alternating-sign matrices

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Abstract

Let $X^{\mathrm{O}}_{2n}(r,t)$ be the generating polynomial of off-diagonally symmetric alternating-sign matrices of order $2n$, where r records the number of nonzero entries strictly above the diagonal and t records the position of the 1 in the first row. Behrend, Fischer and Koutschan conjectured a closed expression for its third boundary coefficient. We prove, for every $n \geq 2$ and as an identity in $\mathbb{Z}[r]$,

$$[t^3]X^{\mathrm{O}}_{2n}(r,t) = r\big(1+(n-2)r^2\big)X^{\mathrm{O}}_{2n-2}(r,1).$$

The source Pfaffian reduces the problem to a relation between two adjacent minors of a skew coefficient matrix. A lower-triangular binomial congruence centres its kernel and makes the resulting matrix bipartite with respect to index parity. Pfaffian minor summation then annihilates the exceptional shifted minor and proves the required relation for all sizes. The result is an exact finite-size boundary identity for the associated six-vertex model; no thermodynamic limit is taken.

Keywords: alternating-sign matrix; six-vertex model; Pfaffian; boundary correlation; exact enumeration

1 Introduction

Alternating-sign matrices (ASMs) and six-vertex partition functions provide a well-developed bridge between exact enumeration and integrable statistical mechanics. Symmetry classes of ASMs lead to modified boundary geometries and to determinant or Pfaffian partition functions [1, 2]. In particular, Behrend, Fischer and Koutschan (BFK) introduced a triangular-domain six-vertex construction for diagonally symmetric ASMs and obtained Pfaffian generating functions for several statistics [3].

The even off-diagonally symmetric specialization has zero diagonal. BFK proved a reflection relation and the coefficient formula at the second admissible boundary position, but stated the corresponding third-position formula for a general weight r as a conjecture [3, Eq. (8.33)]. Refined enumeration at $r = 1$ follows from earlier work [2], while the domino-tiling result of Lee supplies the $r = \sqrt{2}$ specialization [3, 4]. Kumari's subsequent results resolve the three OSASM conjectures from the earlier version of BFK, rather than this new formal-r identity [3, 5].

Here we prove the formal-r conjecture in full. The mechanism has two parts. First, a low-t Laurent expansion of the last column in the BFK Pfaffian reduces the desired coefficient to two adjacent Pfaffian minors. Second, a binomial change of basis centres the bulk kernel. In the centred basis, entries with equal index parity vanish. A parity imbalance therefore kills the only extra term in the Pfaffian minor-summation formula [6].

2 Model and result

An ASM of order m is a matrix with entries in $\{-1, 0, 1\}$ whose nonzero entries alternate in each row and column and whose row and column sums are all 1. For $n \geq 1$, let

$$\mathrm{OSASM}(2n) = \left\{A : \ A \text{ is an ASM of order } 2n, \quad A = A^{\mathsf{T}}, \quad A_{ii} = 0\right\}.$$

For $A \in \mathrm{OSASM}(2n)$, define

$$R(A) = \sum_{1 \leq i < j \leq 2n} |A_{ij}|, \qquad T(A) = \text{the column of the unique 1 in the first row.}$$

Thus $T(A) \in \{2, \ldots, 2n\}$. The refined polynomial is

$$X^{\mathrm{O}}_{2n}(r, t) = \sum_{A \in \mathrm{OSASM}(2n)} r^{R(A)} t^{T(A)} \in \mathbb{Z}[r, t]. \tag{1}$$

Under the BFK bijection, this is a finite triangular-domain six-vertex partition function with a marked boundary position [3]. Both weights in (1) are dimensionless.

Theorem 1. *For every integer $n \geq 2$,*

$$[t^3] X^{\mathrm{O}}_{2n}(r, t) = r(1 + (n-2)r^2) X^{\mathrm{O}}_{2n-2}(r, 1) \qquad \text{in } \mathbb{Z}[r]. \tag{2}$$

Equivalently, if $N_n(\rho, \tau)$ counts the matrices with $(R, T) = (\rho, \tau)$, and $C_n(\rho)$ counts them with $R = \rho$, then

$$N_n(\rho, 3) = C_{n-1}(\rho - 1) + (n-2) C_{n-1}(\rho - 3). \tag{3}$$

Equation (3) is an algebraic consequence of the proof; a direct bijection realizing its two terms is not asserted.

3 Reduction to an adjacent Pfaffian minor

Let $m = 2n$ and $k = n - 1$. Write $A = (a_{ij})_{i,j \geq 0}$ for the skew coefficient matrix with ordinary generating function

$$A(u, v) = \sum_{i,j \geq 0} a_{ij} u^i v^j = \frac{(v - u)(1 + u)(1 + v)}{(1 - uv)\big((1 - ru)(1 - rv) - uv\big)}. \tag{4}$$

This is the coefficient kernel in the BFK Pfaffians [3, Eqs. (8.10)–(8.11)]. For an even-cardinality index set I, A_I denotes the corresponding principal submatrix. Define

$$C_0 = \mathrm{Pf}\, A_{\{0,1,\ldots,2k-1\}}, \qquad C_1 = \mathrm{Pf}\, A_{\{0,1,\ldots,2k-2,2k\}}. \tag{5}$$

Lemma 2. *The BFK Pfaffian formula implies*

$$[t^3] X^{\mathrm{O}}_{2n}(r, t) = r^n\big((1 - r - r^2) C_0 + r C_1\big), \tag{6}$$

$$X^{\mathrm{O}}_{2n-2}(r, 1) = r^{n-1} C_0. \tag{7}$$

Proof. For even m, the non-Pfaffian term in BFK Eq. (8.11) is absent. The interior block of its skew matrix is (4); if $b_i(t)$ is the entry in row i of the last column, then

$$b_i(t) = [u^i] \, \frac{(t - 1 - rtu)(1 + u)}{(rt - (t-1)u)\big(r - (t - 1 + r^2)u\big)}. \tag{8}$$

For $i \geq 1$, expansion of the two denominator factors as geometric series gives the two lowest Laurent coefficients

$$[t^{-i-1}]b_i = (-1)^{i+1} r^{-i-2}, \qquad [t^{-i}]b_i = (-1)^i (i + r + r^2) r^{-i-2}. \tag{9}$$

For the first equality, only the constant term of the numerator and the summand with geometric degrees $(i, 0)$ contribute. At the next power, the contributions have degrees $(i, 0)$, $(i-1, 1)$, and $(i-1, 0)$ from the term linear in u; their sum yields the second coefficient in (9).

Expanding the Pfaffian along its last column, with sign $(-1)^i$ for row i, shows that only $i = m-2, m-3$ can affect $[t^3]$. Equations (5) and (9) give

$$\begin{aligned} [t^{-(m-1)}] \operatorname{Pf} M &= -r^{-m} C_0, \\ [t^{-(m-2)}] \operatorname{Pf} M &= r^{-m}(m - 2 + r + r^2) C_0 - r^{-(m-1)} C_1. \end{aligned} \tag{10}$$

Now BFK Eq. (8.11) reads

$$X_m^{\mathrm{O}}(r, t) = r^{3m/2} t^{m+1} (t-1)^{-(m-1)} \operatorname{Pf} M.$$

Since $m - 1$ is odd,

$$(t-1)^{-(m-1)} = -\big(1 + (m-1)t + O(t^2)\big).$$

Substitution of (10), followed by $m = 2n$, proves (6). The specialization of BFK Eq. (8.10) at order $m - 2$ is exactly (7). □

By Lemma 2, Theorem 1 will follow once we establish

$$C_1 = (1 + kr) C_0. \tag{11}$$

4 Binomial centring of the skew kernel

Remove the two elementary boundary factors from (4):

$$B_r(u, v) = \frac{v - u}{(1 - uv)\big((1 - ru)(1 - rv) - uv\big)}, \qquad A(u, v) = (1 + u)(1 + v) B_r(u, v). \tag{12}$$

For a scalar c, introduce the infinite unit lower-triangular binomial matrix

$$T(c)_{ij} = \begin{cases} \binom{i}{j} c^{i-j}, & i \geq j, \\ 0, & i < j. \end{cases} \tag{13}$$

It satisfies $T(c)T(d) = T(c + d)$. If $K(u, v)$ is the ordinary generating function of a coefficient matrix K, direct summation in (13) gives

$$\operatorname{ogf}\Big(T(c) K T(c)^{\mathsf{T}}\Big) = \frac{1}{(1 - cu)(1 - cv)} K\left(\frac{u}{1 - cu}, \frac{v}{1 - cv}\right). \tag{14}$$

Apply (14) to B_r with $c = -r/2$, and denote the transformed coefficient matrix by E. Rational simplification yields

$$E(u, v) = \frac{v - u}{\big((1 + ru/2)(1 + rv/2) - uv\big)\big((1 - ru/2)(1 - rv/2) - uv\big)}. \tag{15}$$

The two denominator factors in (15) are interchanged by $(u, v) \mapsto (-u, -v)$, whereas the numerator changes sign. Hence

$$E(-u, -v) = -E(u, v), \qquad E_{ij} = 0 \quad \text{if } i + j \text{ is even.} \tag{16}$$

Thus E only couples even indices to odd indices.

Let $S_{i,j} = \delta_{i,j+1}$. Because $T(-r/2)^{-1} = T(r/2)$, restoring the boundary factors in (12) gives

$$A = MEM^{\mathsf{T}}, \qquad M = (I + S)T(r/2). \tag{17}$$

The matrix M is unit lower triangular. Its relevant first-subdiagonal entry is

$$M_{2k,2k-1} = T(r/2)_{2k,2k-1} + T(r/2)_{2k-1,2k-1} = 1 + kr. \tag{18}$$

5 Evaluation of the adjacent minor

For any finite, even-cardinality row set I, Pfaffian minor summation gives

$$\mathrm{Pf}(MEM^{\mathsf{T}})_I = \sum_{\substack{J \subset \mathbb{Z}_{\geq 0} \\ |J| = |I|}} \det(M_{I,J}) \, \mathrm{Pf}(E_J). \tag{19}$$

Although the matrices have been written with infinite indices, the sum in (19) is finite: lower triangularity bounds every column index by the largest row index. Formula (19) may also be obtained directly by expanding the alternating two-form associated with E.

Set

$$I_0 = \{0, \ldots, 2k - 1\}, \qquad I_1 = \{0, \ldots, 2k - 2, 2k\}.$$

For I_0, unit lower triangularity leaves only $J = I_0$, so

$$C_0 = \mathrm{Pf}\, E_{I_0}. \tag{20}$$

For I_1, the first $2k - 1$ rows force the columns $0, \ldots, 2k - 2$. The last column is consequently either $2k - 1$ or $2k$. Equations (17) and (19) therefore give

$$C_1 = M_{2k,2k-1} \, \mathrm{Pf}\, E_{I_0} + \mathrm{Pf}\, E_{I_1}. \tag{21}$$

The set I_1 contains $k + 1$ even indices and $k - 1$ odd indices. By (16), every nonzero entry of E pairs indices of opposite parity. No perfect matching of I_1 can therefore contribute, and

$$\mathrm{Pf}\, E_{I_1} = 0.$$

Using (18) and (20) in (21) proves (11), without division by C_0. Finally, Lemma 2 gives

$$\begin{aligned} [t^3] X^{\mathrm{O}}_{2n}(r, t) &= r^n \Big((1 - r - r^2) + r(1 + (n - 1)r) \Big) C_0 \\ &= r^n (1 + (n - 2)r^2) C_0 \\ &= r(1 + (n - 2)r^2) X^{\mathrm{O}}_{2n-2}(r, 1), \end{aligned}$$

which completes the proof of Theorem 1.

Remark 3. *The Laurent extraction is initially performed over $\mathbb{Q}(r)$, and the binomial congruence over $\mathbb{Q}[r]$. The final equality is between polynomials in $\mathbb{Z}[r]$; it therefore includes $r = 0$ and does not depend on dividing by a possibly vanishing partition function.*

6 Finite-size interpretation and checks

For positive r, the coefficients of (1) are nonnegative Boltzmann weights. Theorem 1 fixes the weighted boundary one-point contribution at $T = 3$ in terms of the unrefined partition function two sizes lower. Coefficientwise, (3) says that the weight-ρ sector receives one shifted copy of the smaller model and $n - 2$ copies shifted by three powers of r. This statement is exact at finite size and under the BFK triangular/reflection boundary conditions.

At the first two sizes the identity reads

$$[t^3]X_4^{\mathrm{O}}(r,t) = r^2, \qquad [t^3]X_6^{\mathrm{O}}(r,t) = 3r^3 + 3r^5,$$

in agreement with direct definition-level enumeration and independent evaluation of the source Pfaffian. Exact integer-polynomial implementations also agree coefficientwise through the reserved order-16 check. These computations test conventions, signs and low-order cases; the all-n conclusion follows from the parity argument above, not from finite extrapolation.

No thermodynamic limit, boundary free energy, phase transition or experimental observable is inferred here. The result concerns the exact finite partition polynomial and its formal weight. The binomial-centred form (15) may nevertheless be useful for other nearby boundary minors, since any centred index set with unequal even and odd cardinalities has vanishing Pfaffian.

7 Conclusion

The formal-r third-boundary identity conjectured by BFK follows from a short structural property of their Pfaffian kernel. Binomial centring exposes an opposite-parity support rule; Pfaffian minor summation then reduces the shifted minor to a single first-subdiagonal entry of the change-of-basis matrix. This proves the conjecture for every $n \geq 2$ while retaining the exact finite-size six-vertex boundary conditions.

Data and code availability

The accompanying reproducibility package contains the exact symbolic checker, two independent finite-size implementations, coefficient data and the commands used to regenerate the reported checks. No experimental or personal data are involved.

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OpenAI Codex (model `gpt-5.6-sol`, reasoning effort `ultra`) was used to assist with literature discovery, symbolic derivation, exact-computation code, independent checking in a separate model context, and generation and revision of the manuscript text. Its outputs, the cited sources and the underlying calculations require critical review by the author before submission; the AI system is not an author and cannot assume responsibility for the work.

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